SHAHRAM AJLOO

BMS, utilities and validation · Improvements in an aseptic filling plant

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Novo Nordisk A/S · Site Bagsværd, DPM BA · BMS Supporter (Adv. Utilities Professional)

Application: BMS Supporter — Facility and Utility Projects

Dear Ehab El-sayed

In an aseptic filling factory the utilities are not building services. WFI, clean steam, compressed air and HVAC are the conditions that make a batch valid, and a change in the BMS is therefore a GMP change before it is an engineering one. That is the part of this role I find worth doing: one line in a specification becomes a thousand details on the floor — tag lists, alarm limits, I/O checks, calibration, requalification — and it all has to be right before the line restarts.

I would work the projects from four fixed points. First, plan backwards from the production stop. The two annual stops are the deadline, not the start, so everything that can be prepared is prepared before the factory stops: user requirements, design review, FAT, contractor access, spare parts, test scripts and permits. During the stop there should be execution and verification left — nothing else, because the stop has no slack. Second, every alarm must have a reason: what is monitored, at which limit, with what consequence and who acts on it. A cleanroom where the alarm list has grown for ten years hides the one signal that matters, and alarm rationalisation is one of the cheapest quality improvements available. Third, validation is written alongside the change, not afterwards — requirements, risk assessment, IQ/OQ/PQ and a traceability matrix that can be read by someone who was not in the room. Fourth, contractors on a live plant: one coordination meeting a day during the stop, one plan, one accountable person per scope, and access, permits and cleanroom behaviour agreed before anyone arrives.

I have done this work in your own organisation. From 2022 to 2024 I was at Novo Nordisk as senior commissioning engineer and project manager on electrical systems, with 37 parallel capital projects and 25 specialists in Kalundborg and Bagsværd: documented acceptance criteria, FAT and SAT, IQ/OQ/PQ, deviation handling and audited handover with zero critical findings. Before that, five years of maintenance and operations on a 24/7 plant at Biofos — pumps, HVAC, high-voltage switchgear and SCADA-based condition monitoring, LOTO, APV, near-miss routines and a PDCA programme that took 40 % off the energy cost. That is where I learned that a control system is only as good as the sensor loop behind it, and that an alarm nobody trusts is worse than no alarm at all.

What I would bring to Facility and Utility Projects is the combination of the three disciplines this role sits between. I am an authorised marine engineer with electrical authorisation and a B.Tech in Technology Management and Mechanical Engineering, so I can read the P&ID and the electrical drawing as well as the automation specification. I have run projects in a GMP environment where the documentation is part of the deliverable, not an appendix. And I am used to speaking to both the automation engineer and the skilled worker on the floor — the same message, in the language each of them needs, which is most of what coordination actually is.

I know the systems this role sits in. At Novo Nordisk I worked in the BMS and FMS in daily operation and through projects — alarm and point lists, changes to limits and graphics, the verification that follows a change, and the documentation that has to hold when QA or an auditor reads it. I am confident in the system and comfortable being the one operations call when a signal does not behave. What I add around it is the discipline: preparation before a stop, a reason for every alarm, and validation written while the change is still being designed.

What draws me back to Novo Nordisk is that the purpose is measurable. The products made in Bagsværd — insulin, GLP-1 and weight-management medicines — are the reason a utility project matters to patients, and the Novo Nordisk Way is unusually concrete about it: we never compromise on quality and business ethics, we are open and honest, and we speak up. That last part is not a slogan in a factory that cannot stand still — it is the mechanism that stops a small deviation from becoming a stopped line, and it fits the way I work anyway.

Practically: I live in Måløv, about ten minutes from Bagsværd. I speak Danish natively and English professionally, written and spoken, which is the working reality in an international organisation. I am available for both production stops, including the July/August and December/January periods, and I am used to shutdown work from 24/7 operations. I am also an active Claude coder and build my own Python tools for documentation and follow-up — traceability matrices, test and deviation logs, and investment cases with NPV, IRR and scenarios when equipment is to be replaced.

I would like to come to Bagsværd, walk the utility areas with you and your team, and talk through how I would prepare the next stop — and where I think the quickest quality gains are in the alarm and documentation setup.

Kind regards

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PROFILE

Project and commissioning engineer with 20+ years from technical operations, regulated production and utilities — authorised marine engineer (maskinmester) with electrical authorisation and a B.Tech in Technology Management and Mechanical Engineering. Two years at Novo Nordisk with 37 parallel capital projects under GMP/FDA/EMA, FAT/SAT and IQ/OQ/PQ, zero critical findings at handover; five years of maintenance and operations on a 24/7 plant with SCADA, HVAC, pumps and high-voltage switchgear. Works hands-on in the BMS and FMS, from requirement to shut-down plan.

UTILITIES, VALIDATION AND RESULTS

- **Validation and qualification:** FAT/SAT and IQ/OQ/PQ across 37 capital projects with documented acceptance criteria, deviation handling and audited handover; zero critical findings. Traceability from requirement to test, written for the next reader rather than for the auditor.
- **BMS, FMS and control systems:** Two years in the BMS and FMS at Novo Nordisk — alarm and point lists, limit and graphic changes, and the verification after them — plus five years of SCADA monitoring and alarm handling on a 24/7 plant.
- **Shutdown and stop planning:** Planned and executed work in 24/7 operations with a fixed window: preparation, permits, contractors, test scripts and verification sequenced so the restart is not delayed.
- **Contractors and stakeholders:** Coordinated external contractors, suppliers and internal operations across 37 projects and 11 construction projects; one plan, named owners and a daily coordination point.
- **Improvement with measured effect:** PDCA, 5S, standard work, A3 and 5 Whys in daily use. A PDCA programme cut energy cost by 40 %; own Python/Claude tools have taken over 40 % of the administrative work out of documentation and follow-up.

TOOLS AND METHODS

Task	Tool and documented use
Validation and change control	URS, risk assessment, IQ/OQ/PQ, traceability matrix and deviation handling; change control before anything is altered, and acceptance documented at FAT/SAT.
Automation and alarms	BMS/FMS, SCADA and PLC-based systems, sensor loops, alarm limits and escalation; the alarm list treated as a quality document.
Project management in production	Stage plans, shutdown schedules, test scripts and readiness checks; daily coordination with operations and contractors.
Technical documentation	P&ID and electrical drawings, functional descriptions, as-built documentation and handover packages; decisions written with owner and deadline.
Safety	LOTO, APV/workplace assessment, risk assessment, permits and near-miss reporting; contractor safety briefings before work starts in a production area.
Improvement and data	PDCA, 5S, standard work, VSM, A3 and 5 Whys with verified countermeasures; KPI follow-up and investment cases with NPV, IRR, sensitivity and scenarios.

EXPERIENCE

Senior Commissioning Engineer / Project Manager, electrical systems — Novo Nordisk A/S · Kalundborg / Bagsværd, Denmark · Nov 2022–Jul 2024

Scope: 37 capital projects · 25 specialists · DKK 2–10m each · GMP/FDA/EMA

- **Qualification and documentation:** FAT/SAT and IQ/OQ/PQ with traceable documentation and deviation handling; zero critical findings at handover.
- **BMS and FMS changes:** Worked in the BMS and FMS in daily operation and in projects: alarm and point lists, limit and graphic changes, verification after change and the validation documentation that follows.
- **Coordination and reporting:** Progress, cost, risk and resources across engineering, procurement and operations; monthly reporting to management.

Owner / Technical Director — ELOHVAC ApS · Copenhagen, Denmark · May 2023–present

Scope: 5 electricians and 3 apprentices · full P&L · commercial and industrial customers

- **Commercial and operational ownership:** Quoting, pricing, margin follow-up and service agreements with response time, quality and delivery as KPIs.
- **People and safety:** Daily planning, 1:1s, reviews, competence matrix and apprentice training; LOTO, risk assessment and APV at every site start.
- **Digital tools:** Python and Claude API for quotations, schedules, checklists and KPI dashboards; 40 % productivity gain.

Project and Workshop Manager — Ventek-Air · Copenhagen, Denmark · Jan 2022–Nov 2022

Scope: 14 employees · 11 parallel construction projects · DKK 2–10m each

- **Throughput and flow:** Tenders, quotations, subcontractors and handover packages; daily board meetings and bottleneck analysis lifted delivery performance.

Maintenance and Operations Manager / Workshop Technician — Biofos, Lynetten · Copenhagen, Denmark · Aug 2017–Jan 2022

Scope: 24/7 plant · pumps, HVAC, switchgear, SCADA · shifts including nights and weekends

- **Operation and maintenance:** Preventive maintenance planned from SCADA condition data; alarm handling, fault diagnosis and corrective work prioritised by consequence.
- **Safety and standards:** LOTO, risk assessment, APV and near-miss reporting in shift operations; standard work, 5S and PDCA on the workshop and stores.

Electromechanical Technician / Production Team Leader — ARJ Engineering · Tehran, Iran · Dec 2003–Mar 2010

Scope: 12 mechanics and technicians · machine workshop · industrial equipment

- **Equipment and targets:** Shift planning, material control and target follow-up; fault diagnosis, root-cause analysis and corrective actions on industrial machinery.

EDUCATION AND CERTIFICATION

B.Tech, Technology Management & Mechanical Engineering · Copenhagen School of Marine Engineering & Technology Management · 2017–2022 · EQF 6 · GPA 7.6/12

Authorised marine and electrical engineer (LV/MV/HV) · Danish Safety Technology Authority · May 2023

Electrical and mechanical design, electrical engineering · Tehran Technical University · 2000–2003 · EQF 6

Refrigeration and air conditioning · STCW safety and engine-room watch · Mercantec 2021 · Danish Maritime Authority 2021–2023

LANGUAGES, PRACTICAL AND REFERENCES

Danish: native · English: professional, written and spoken · Persian: native · Driving licence B · SAP and Excel in daily use · Lives in Måløv, about ten minutes from Site Bagsværd · Available for both annual production stops, including July/August and December/January · References from Novo Nordisk and Biofos on request.

WHAT THE ROLE ASKS FOR — MY EVIDENCE

Requirement	Evidence
Relevant technical background, e.g. mechanical engineering	Authorised marine engineer (maskinmester) with electrical authorisation LV/MV/HV and B.Tech in Technology Management and Mechanical Engineering (EQF 6).
Strong Danish and English, international environment	Danish native, English professional written and spoken, Persian native; two years in an international project organisation with English as the working language.
Communication and collaboration at all levels of the organisation	Coordinated 25 specialists, operations, procurement, contractors and management across 37 projects; written decisions with owner and deadline.
Proactive and ambitious, flexible during the two production stops	Five years of 24/7 operations and shutdown work; planned and executed work in fixed windows where the restart could not slip. Available for July/August and December/January.
Validation of systems (useful)	FAT/SAT and IQ/OQ/PQ across 37 projects with acceptance criteria, risk assessment and FMEA, deviation handling and audited handover; zero critical findings.
BMS/FMS experience (useful)	Two years working in Novo Nordisk's BMS and FMS: alarm and point lists, limit and graphic changes, verification after change and validation documentation, plus five years of SCADA-based monitoring and alarm handling on a 24/7 plant.
Pharmaceutical industry experience (useful)	Two years at Novo Nordisk in a GMP/FDA/EMA environment, with documentation, change control, compliance and audit readiness as daily conditions.
Driving changes from specification to implementation	Delivered 37 capital projects and 11 construction projects from requirements through FAT/SAT to handover, with contractors and operations coordinated to a fixed date.

FIRST 90 DAYS — SYSTEM, STOP, DOCUMENTATION

Period	What happens, and how it is measured
Week 1–2	Walk the utility areas with the engineers and skilled workers: WFI, clean steam, compressed air, HVAC and the plant rooms. Sit with the BMS in daily operation — alarm list, active alarms, history and the points that are bypassed. Read the last two stops' documentation and the open deviations.
Day 30	One-page baseline: the twenty most frequent alarms and what they actually mean, open changes in the system, the age of the documentation, and where the maintenance and requalification plans have drifted. 1:1 with each colleague I will depend on.
Day 60	First alarm review completed with an agreed list of signals to remove, re-limit or re-route. Validation template and traceability matrix for BMS changes agreed with QA. Contractor plan and readiness checklist for the next stop drafted with the team.
Day 90	Next production stop planned to the hour with test scripts, permits, access and named owners. Two improvements delivered and documented — one in the system, one in the routine — and a written handover of the way I work, so it does not depend on me being in the building.

WHY THE UTILITIES ARE THE PROCESS

A cleanroom is not clean because it was built that way; it is clean because the air handling, the pressure cascades and the monitoring behind them keep doing what they were qualified to do. The same is true of WFI and clean steam: the specification lives in a control loop, a sensor and an alarm limit, and when one drifts, the batch is at risk long before anyone sees a deviation. So I treat the monitoring system as part of the product, not part of the building.

PLANNING BACKWARDS FROM THE STOP

A stop is a project with a fixed end date and no possibility of extension, so the plan is built back from the restart. What is left for the stop itself is execution and verification — installation, loop checks, functional tests, documentation and release. Everything else belongs before it: design review, FAT, spare parts on site, contractor induction, permits, area access, and a test script read by someone other than its author. Twice a year is a rhythm, and a rhythm can be prepared.

ALARMS, SENSORS AND DATA INTEGRITY

An alarm is a decision someone has to make at three in the morning. If there are four hundred of them, nobody makes it. So the alarm list is a quality document: what is critical, what is informative, what is a maintenance message dressed as an alarm, and which signals were muted so long ago that nobody remembers why. Behind every alarm sits a sensor, and behind every sensor a calibration, a trend and an audit trail. I have worked in the BMS and FMS myself, so the alarm list is not a black box to me: I know what a change to a limit or a graphic does downstream, and what has to be verified afterwards.

VALIDATION AND CHANGE CONTROL

The cheapest change is the one documented while it is designed. I write requirements, risk assessment, test cases and the traceability matrix as the project moves, not at the end, because that is the only way the documentation survives a change of scope. Change control comes before the work; deviations are raised when they happen, not when they surface at handover; and acceptance criteria are agreed with the people who will run the plant afterwards.

CONTRACTORS ON A LIVE PLANT

External specialists are necessary and expensive, and on a live aseptic plant they are also the biggest variable. So: one plan for everyone, one coordination meeting a day during execution, one accountable person per scope, and access, permits, LOTO, cleanroom behaviour and waste handling agreed before the first toolbox talk. Contractors who know what is expected of them deliver faster than contractors managed by phone.

LEADING WITHOUT A TEAM

This role has no direct reports, and that changes what leadership is. It becomes the quality of the one-page description someone else works from, the speed with which QA and operations hear about a change before it happens, and the willingness to say that a plan will not hold before the stop rather than after. I have led 25 specialists and 12 technicians with formal responsibility, and I have run projects where my only authority was the plan and the credibility behind it.

DIGITAL WORK AND AI

I am an active Claude coder and build my own tools in Python: traceability matrices that stay in step with the test list, deviation and punch-list logs filtered by area and owner, alarm statistics pulled into one page, and investment cases with NPV, IRR, sensitivity and scenarios when equipment is replaced. None of it replaces the validated systems; it shortens the documentation work.